

ASCII MASTER FLOW – PANEL 1/12: Property structure processing map

COMPOSITION + BONDING -> intrinsic behaviour

CRYSTAL / MOLECULAR STRUCTURE -> property pathways

MICROSTRUCTURE + DEFECTS -> local performance

PROCESSING -> changes structure and defects

TEST -> strength != stiffness != toughness != hardness

ASCII MASTER FLOW – PANEL 2/12: Advanced materials family

ADVANCED MATERIAL -> engineered demanding function

COMPOSITE -> matrix + reinforcement

CARBON-FIBRE COMPOSITE -> high specific strength/stiffness

SMART MATERIAL -> stimulus-responsive property

EXAMPLES -> shape-memory alloy | piezoelectric material

ASCII MASTER FLOW – PANEL 3/12: REE classification rail

REE = 15 LANTHANIDES + SCANDIUM + YTTRIUM

LIGHT -> La | Ce | Pr | Nd

HEAVY OWNER EXAMPLES -> Dy | Tb | Y

MONAZITE -> chiefly light-REE-rich Indian context

TRAP -> geological occurrence != separated magnet input

ASCII MASTER FLOW – PANEL 4/12: Category firewall

CHEMICAL GROUP -> rare earth elements

POLICY CATEGORY -> critical minerals

STATUTORY ROUTE -> Part D

EXPLORATION ROUTE -> Seventh Schedule

LITHIUM / COBALT / NICKEL / GRAPHITE -> critical may be, REE no

ASCII MASTER FLOW – PANEL 5/12: Permanent magnet capability ladder

ORE -> SEPARATION -> OXIDE / METAL

METAL -> ALLOY -> MAGNET

MAGNET -> MOTOR / ELECTRONICS / DEFENCE

Nd -> magnetic performance

Dy / Tb -> high-temperature qualification

ASCII MASTER FLOW – PANEL 6/12: Battery mineral role matrix

LITHIUM -> charge-carrier system input

NICKEL + COBALT -> NMC / NCA cathodes

GRAPHITE -> common lithium-ion anode

LFP -> neither nickel nor cobalt

SODIUM-ION -> avoids lithium

ASCII MASTER FLOW – PANEL 7/12: Named application portfolio

GALLIUM / GERMANIUM -> high-tech electronics

COPPER -> electrical systems

REE -> magnets | phosphors | catalysts | specialised optics

TITANIUM-BEARING MINERALS -> separate policy / geography input

MONAZITE -> rare-earth values + thorium-bearing context

ASCII MASTER FLOW – PANEL 8/12: Mine to technology value chain

EXPLORE -> ASSESS -> DEVELOP -> EXTRACT

BENEFICIATE -> SEPARATE / REFINES

OXIDE / METAL / ALLOY -> COMPONENT

COMPONENT -> BATTERY / MAGNET / CHIP / SYSTEM

RECYCLE -> recovered feed with purity/economic limits

ASCII MASTER FLOW – PANEL 9/12: Dependency heat map

UPSTREAM -> geology | mine | permitting
MIDSTREAM -> separation | refining | intermediate
DOWNSTREAM -> cell | magnet | chip | equipment
ENABLERS -> technology | finance | logistics | offtake
VERDICT -> secure mine alone is partial security

ASCII MASTER FLOW – PANEL 10/12: Circular materials loop

DESIGN -> durability | repair | disassembly

USE -> reuse / remanufacture

COLLECT -> sort -> pre-process

RECOVER -> purify -> re-enter manufacture

SUBSTITUTE + EFFICIENCY -> reduce primary-material pressure

MUST REMEMBER: Critical-mineral analysis follows geology, resource/reserve, mining,...

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ASCII MASTER FLOW – PANEL 11/12: Mine-life safeguard chain

SITING -> cumulative impact + consultation

OPERATIONS -> dust | noise | water | tailings controls

COMMUNITY -> FRA due diligence | disclosure | grievance

RECLAMATION -> topsoil | stable dumps | native restoration

CLOSURE -> finance + monitoring; plan != achieved outcome

CLOSE DISTINCTION: Critical is not necessarily rare, rare earths are not all scarce,...

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ASCII MASTER FLOW – PANEL 12/12: India institutions status and PYQ rail

MINES -> policy | GSI -> survey/explore | IBM -> plans/conservation

STATE GRANT + s.11D CENTRAL AUCTION -> distinct roles

IREL / DAE -> monazite route | KABIL -> overseas route

MoES / NIOT / ISA -> deep-ocean exploration boundary

REPORT -> LAW -> APPROVAL -> MOU -> EXPLORATION -> PRODUCTION -> OUTCOME

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name mineral/material, value-chain node,...

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